Colorado Motor Speech Framework

Assess	*Task*	*Characteristics*
Rate of Speech	***Complete running speech tasks, which may include:*** ***1) Conversational speech:*** *Ask questions such as, "What does a typical day look like for you?"* ***2) Narrative task:*** *Ask the patient to describe a complex picture (e.g.,. Kite Picture, Cookie Theft Picture).* ***3) Oral reading task:*** *Ask the patient to read a standardized passage aloud (e.g., Grandfather Passage, Caterpillar Passage).*	Slow rate of speech Fast rate of speech Variable rate of speech Short phrases
Fluency and Cadence		Abnormal noises which interrupt speech or occur when pt is not speaking Stutter-like disfluencies (sound/syllable repetition) Distorted substitutions or articulatory additions Syllable segregation Prolonged interword intervals Prolonged phonemes
Vocal Loudness		Reduced vocal loudness (<70dB) Explosive loudness bursts (often at the start of a phrase) Monoloudness Loudness decay (per utterance) Excessive loudness variation
Voice Quality and Respiration	***1) Observe during running speech tasks.*** ***2) Measure maximum phonation time:*** *Instruct the patient, "Take a deep breath and hold out the sound 'ah' for as long as you can, until you run out of air."* *- MPT < 15 seconds is considered atypical in healthy adults* *- MPT <12 seconds is considered atypical in healthy older adults ages 65+*	Breathiness Aphonia Hoarseness *(note: may have non-neurological causes)* Strain Diplophonia (presence of two pitches) Pitch breaks Monopitch Pitch Instability Voice stoppage Maximum phonation time <15 seconds for adults under age 65, <12 seconds for adults over age 65 Runs out of air when speaking Vocal tremor Rapid vocal flutter (i.e. rapid voice tremor)
Word and Phrasal Stress	***1) Observe during running speech tasks.*** ***2) Stress-marking task:*** *Present patient with a sentence in writing, such as, "He went grocery shopping yesterday." Ask questions to determine whether the patient can produce different phrasal stress patterns. For example:* *Who went grocery shopping? (HE went grocery shopping.)* *He went shopping today? (He went shopping YESTERDAY).*	Reduced use of stress (i.e. minimal emphasis placed on any word or syllable; may impact how meaning is conveyed or implied) Prosodic excess or scanning (i.e. equal emphasis on each syllable or word; may have word-by-word cadence) Atypical silences / pauses Errors marking stress (i.e. emphasis placed on the wrong word or syllable)

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Assess	*Task*	*Characteristics*
Articulation	***1) Observe during running speech tasks.*** ***2) Word-level articulation task:*** *Repeat these words after me: northwest, twist, fused, sleeps, scarecrow, playground, lifeboat, shipwreck, mushroom, woodwork." (from Tikofsky, 1970)* ***3) Sentence-level articulation task:*** *Ask the patient to either read the following sentences aloud or repeat them after you:* *- We saw several wild animals.* *- My physician wrote out a prescription.* *- The municiple judge sentenced the criminal.* ***4) Assess automatic speech:*** *"Tell me the days of the*	Imprecise consonants *Note if errors occur consistently based on place / manner of articulation (i.e. lingual phonemes, oral pressure consonants, etc.)* Distorted Vowels Irregular articulatory breakdown Articulatory groping (i.e. movement of the jaw, tongue, or lips in an attempt to correctly place for speech) Telescoping (i.e. syllable series run together or multisyllabic words are collapsed) Deterioration of speech during continuous speaking Fewer errors in automatic speech than connected speech
Resonance	***1) Observe in running speech tasks.*** ***2) Use nasal occlusion to evaluate resonance and nasal air emission:*** *"Repeat after me: 'Buy Bobby a puppy.' Now, pinch your nostrils shut. Again, 'Buy Bobby a puppy.'" Compare resonance when the nares are not occluded versus occluded.*	Audible nasal emission or nasal snorting Hyponasality Hypernasality
Diadochokinetic Rate (DDK)	***1) AMR: "Take a breath and repeat 'puh-puh-puh-puh-puh as fast and as steadily as you can, until I tell you to stop."*** *Provide a 2- to 3-second example. Repeat with "tuh" and "kuh."* ***2) SMR: "Now, take a breath and repeat 'puh-tuh'kuh' as fast and as steadily as I can, until I tell you to stop."*** *Provide a 2- to 3-second example. If this speech sequence cannot be performed, ask the patient to repeat "buttercup, buttercup, buttercup."*	Slow rate: *AMR= <5-7 repetitions per second* *SMR= <5 repetitions per second* Irregular rhythm Rapid, accelerated AMRs Overall imprecise articulation of AMRs / SMRs "Puh" is more imprecise/slow than "tuh" or "kuh" "Tuh" and "Kuh" more imprecise/slow than "Puh" "Puh-tuh-kuh" is poorly sequenced
Self-Report	***Ask the patient:*** *"On a scale of 1-10, 1 being the worst and 10 being the best, how would you rate your speech right now?"*	_____ / 10
Intelligibility	***Judge during running speech tasks.*** *Estimate of the percentage of words correctly understood. If intelligibility is of concern, consider using the Functional Importance to Intelligibility (FITI) Modules (Gurevich & Kim, 2024) for further assessment.*	_____%
Naturalness	***Judge during running speech tasks.*** *How well does speech matches normal stands of rate, pitch, and loudness?*	VAS: |----------|
Efficiency	***Judge during running speech tasks.*** *How efficient are messages conveyed (intelligible words per minute)?*	VAS: |----------|

Colorado Motor Speech Framework

Assess	Task	Characteristics
Emotional Expression	***Observe the patient's emotional state throughout the assessment.*** *Note patient or caregiver report of uncontrolled or inappropriate emotional responses.*	Pseudobulbar Affect (i.e. emotional lability)
		Flat affect/masked facies
General Oral Mech Observations	***Observe across oral mechanism tasks.***	Incoordination / dysmetria
		Muscle atrophy *Location(s):*
		Involuntary, unpredictable movements (lips, jaw, tongue, eyes, etc).
		Tremor: action-induced *Location(s):*
		Tremor: with sustained postures *Location(s):*
		Tremor: at rest *Location(s):*
Overall Face	***Look at the jaw and lips at rest.*** *"Relax , look straight ahead, and breathe comfortably"*	Drooping on entire side of face *L / R*
		Drooping on lower side of face *L / R*
		Bilateral facial weakness
		Hypotonia (i.e. reduced muscle tone)
		Muscle weakness
		Spasticity
		Drooling
		Decreased sensation of the face, oral cavity, tongue, or palate
		Nonverbal oral apraxia (i.e. inability to perform or imitate voluntary oral movements, e.g., "cough" or "click your tongue")
Lips	***1) Retraction:*** *"Smile and hold for three seconds."* ***2) Protrusion:*** *"Pucker your lips."* ***3) Resistance:*** *- During smile/pucker: "Don't let me move your lips."* *- "Squeeze your lips tightly together; don't let me open them."*	Asymmetrical movement *Reduced on L / R*
		Weakness
		Limited range of motion

Colorado Motor Speech Framework

Assess	*Task*	*Characteristics*
Jaw	***1) Strength:*** - *"Open your mouth wide and don't let me close it."* - *"Now close your mouth tightly and don't let me open it."* ***2) Visualize/palpate masseter contraction:*** *"Bite down five times, like you're chewing gum."*	Jaw hangs open at rest Jaw deviation to one side *L / R* Weakness when opening or closing Clonus (shivers when relaxed after clenching the teeth) Limited range of motion Progressive weakening of jaw movements
Soft Palate / Velpoharynx	***1) Adequacy of velar and labial seals:*** *"Puff your cheeks up with air. Don't let me push any air out."* ***2) Symmetry & ROM:*** *"Open your mouth wide. Now say, 'ahhh'."* *--> visualize velum w/ pen light, use tongue blade if needed*	Palate hangs low in mouth Palate asymmetry *Hangs lower on L / R (NOTE slight asymmetry is normal)* Asymmetric movement on phonation Air escapes from nose when puffing cheeks
Tongue	***1) Protrusion:*** *"Stick your tongue out. Hold for 3 seconds."* ***2) Strength:*** *"Don't let me move your tongue." (using tongue blade)* ***3) ROM:*** - *"Curl your tongue up toward your nose, then down toward your chin."* - *"Move your tongue from side to side."* ***4) Lateral strength:*** *"Push your tongue into the inside of your cheek against my finger."*	Fasciculations (i.e. involuntary, fast, spontaneous muscle contraction and relaxation): assess at rest and with protrusion Deviation to one side on protrusion *L / R* Weakness *Bilateral / L / R* Limited range of motion *Bilateral / L / R* Diminished ability to curl tip of tongue Slow movements
Larynx	***1) Instruct patient to cough.*** *Consider adequacy of cough phases: inhalation, compression, expulsion.* ***2) Elicit glottal coup:*** *"Say 'uh-oh.'"*	Weak Cough* Weak Glottal Coup* (i.e. glottal stop or glottal attack) Inhalatory stridor *weak cough but strong glottal coup *weak coup and weak cough

Colorado Motor Speech Framework

Assess	*Task*	*Characteristics*

Colorado Motor Speech Framework: Scoring Form

			Lower Motor Neuron	Upper Motor Neuron		Basal Ganglia		Cerebellum	Left Hemisphere
	Characteristics	***Y/N***	**Flaccid**	**UUMN**	**Spastic**	**Hypokinetic**	**Hyperkinetic**	**Ataxic**	**Apraxia of Speech**
Rate of Speech	Slow rate of speech	☐		X (un-common)	XX		X	X	X
	Fast rate of speech	☐	-	-	-	XX (short rushes of speech)		-	-
	Variable rate of speech	☐				X	X (unpredictable movement)	X	
	Short phrases	☐	XX		X		X		X
Fluency & Cadence	Abnormal noises	☐	-	-	-	-	XX	-	-
	Stutter-like disfluencies	☐				XX			X
	Distorted substitutions or articulatory additions	☐							XX
	Syllable segregation	☐						X	XX
	Prolonged interword intervals	☐				X	XX		X
	Prolonged phonemes	☐					X	X	X
Loudness	Reduced vocal loudness (<70dB)	☐	X (CN X)	X		XX			
	Explosive loudness bursts	☐						XX	
	Monoloudness	☐	X		XX	XX	X	X	X
	Loudness decay	☐				X		X	
	Excessive loudness variation	☐					XX	XX	
Voice Quality and Respiration	Breathiness	☐	XX (CN X or phrenic nerve)			X	X (transient)		
	Aphonia	☐	XX (CN X)		X (intermittent)		X (intermittent)		
	Hoarseness	☐	X (CN X)						
	Strain	☐	X (ex. compensatory for diaphragmatic weakness)	X (un-common)	XX		X (ex. adductor spasmodic dysphonia)	-	
	Diplophonia	☐	XX (CN X)	-	-	-	-	-	-
	Pitch breaks	☐	X		X		X		
	Monopitch	☐	X		XX	XX	X	X	X
	Pitch instability	☐	X			X	X	XX	
	Voice stoppage	☐					XX		
	MPT <15 sec under age 65, <12 sec over age 65	☐	X (CN X or phrenic nerve)		X	X	X		
	Runs out of air when speaking	☐	X		X	X		X	
	Vocal tremor	☐					XX	X	
	Rapid vocal flutter (i.e. rapid voice tremor)	☐	X (CN X)			X			
Word & Phrasal	Reduced use of stress	☐			X	XX			
	Prosodic excess or scanning	☐			X			XX	X
	Atypical silences / pauses	☐				XX	X	X	X
	Errors marking stress	☐							XX

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			Lower Motor Neuron	Upper Motor Neuron		Basal Ganglia		Cerebellum	Left Hemisphere
	Characteristics	***Y/N***	**Flaccid**	**UUMN**	**Spastic**	**Hypokinetic**	**Hyperkinetic**	**Ataxic**	**Apraxia of Speech**
Articulation	Imprecise consonants	☐	X (errors are typically consistent; errors reflect CNs affected)	X	X	X	X	X (inconsistent breakdowns)	X (inconsistent; fewer errors for automatic speech)
	Distorted Vowels	☐	X (CN V, VII, and XII)		X		XX	XX	X
	Irregular articulatory breakdown	☐		X (related to unilateral weakness)			X (due to unpredictable movement)	XX (due to incoordination)	X
	Articulatory groping	☐							XX
	Telescoping	☐						XX (due to poor coordination)	X (due to imprecise articulation)
	Deterioration of speech during continuous speaking	☐	XX (ex. Myasthenia Gravis: improvement after rest)	-	-	-	-	-	-
	Fewer errors in automatic speech	☐	-	-	-	-	-	-	XX
Resonanc	Audible nasal emission or nasal snorting	☐	XX		-	-	-	-	-
	Hyponasality	☐					X (intermittent)	X (intermittent)	
	Hypernasality	☐	XX (CN X)	X	X	X	X (intermittent)		
Diadochokinetic Rate (DDK)	Slow rate	☐	X (may be compensatory for poor articulation)	X	XX (slow and regular AMR)	X	X	X	X
	Irregular rhythm	☐				X	XX (from unpredictable movement)	XX (irregular AMR)	X (SMRs likely more irregular than AMRs)
	Rapid, accelerated AMRs	☐				XX			
	Overall imprecise articulation of AMRs / SMRs	☐	X	X	X	X			X (SMRs likely less precise than AMRs)
	"Puh" is more imprecise/slow than "tuh" or "kuh"	☐	X (CN VII)						
	"Tuh" and "Kuh" more imprecise/slow than "Puh"	☐	X (CN XII)						
	"Puh-tuh-kuh" is poorly sequenced	☐						X	XX
	Self-Rating: ____ / 10	**(select)**							
	Intelligibility: _____%	**(select)**							
	Naturalness:	**(select)**							
	Effiency:	**(select)**							

Colorado Motor Speech Framework: Scoring Form

			Lower Motor Neuron	Upper Motor Neuron		Basal Ganglia		Cerebellum	Left Hemisphere
	Characteristics	***Y/N***	**Flaccid**	**UUMN**	**Spastic**	**Hypokinetic**	**Hyperkinetic**	**Ataxic**	**Apraxia of Speech**
	Pseudobulbar Affect	☐	-	X	XX	X	X	-	-
	Flat affect/masked facies	☐	-	-	-	XX	-	-	-
General OME	Incoordination / dysmetria	☐						XX	
	Muscle atrophy	☐	XX	-	-	-	-	-	-
	Involuntary, unpredictable movements	☐	-	-	-	-	XX	-	-
	Tremor: action-induced	☐				X	XX (Essential Tremor)	X (terminal tremor)	
	Tremor: with sustained postures	☐				X	X		
	Tremor: at rest	☐	-	-	X	XX		-	-
OME (Face)	Drooping on entire side of face	☐	X (CN VII, ipsilateral to lesion)						
	Drooping on lower side of face	☐		XX (droop contralateral to lesion)	XX				
	Bilateral facial weakness	☐	X		X				
	Hypotonia	☐	XX					X	
	Muscle weakness	☐	XX	X	X				
	Spasticity	☐		X	X				
	Drooling	☐	X	X	X	X			
	Decreased sensation	☐	X (note which CN is affected)						
	Nonverbal oral apraxia	☐							XX
OME (Lips)	Asymmetrical movement	☐	X (CN VII; reduced movement ipsilateral to lesion)	X (reduced movement contralateral to lesion)					
	Weakness	☐	X (CN VII; weakness on the impaired side, ipsilateral to the site of lesion)	X (weakness, contralateral to the site of lesion)				-	
	Limited range of motion	☐	X	X (unilateral)	X	X			

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			Lower Motor Neuron	Upper Motor Neuron		Basal Ganglia		Cerebellum	Left Hemisphere
	Characteristics	***Y/N***	**Flaccid**	**UUMN**	**Spastic**	**Hypokinetic**	**Hyperkinetic**	**Ataxic**	**Apraxia of Speech**
OME (Jaw)	Jaw hangs open at rest	☐	X (CN V bilaterally)						
	Jaw deviation to one side	☐	X (CN V unilaterally; jaw deviates to weak side)						
	Weakness when opening or closing	☐	X (CN V)						
	Clonus	☐			X				
	Limited range of motion	☐	X			X			
	Progressive weakening of jaw movements	☐	X (associated with Myasthenia Gravis)						
OME (Soft Palate)	Palate hangs low in mouth	☐	X (CN X)						
	Palate asymmetry	☐	X (CN X; hangs lower on side of lesion)	X (un-common)					
	Asymmetric movement on phonation	☐	X (CN X; palate moves towards nonimpaired side)	X (un-common)					
	Air escapes from nose when puffing cheeks	☐	X (CN X)		X				
OME (Tongue)	Fasciculations	☐	XX (CN XII)						
	Deviation to one side on protrusion	☐	X (CN XII; deviates to weak side, ipsilateral to lesion)	X (deviates to weak side, contralateral to lesion)					
	Weakness	☐	X (CN XII)	X (unilateral)	X				X (commonly right lingual weakness)
	Limited range of motion	☐	X (CN XII)		X	X			
	Diminished ability to curl tip of tongue	☐	X (CN XII)						
	Slow movements	☐	X (CN XII)	X	X			X	
OME (Larynx)	Weak Cough*	☐	X (CN X)			X			
	Weak Glottal Coup*	☐	X (CN X)			X			
	Inhalatory stridor	☐	X (CN X)				X (usually accompanied by hyperkinesias)		
	*weak cough but strong glottal coup	☐	could indicate respiratory weakness						
	*weak coup and weak cough	☐	could indicate laryngeal weakness			X			

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Characteristics	*Y/N*	Lower Motor Neuron	Upper Motor Neuron		Basal Ganglia		Cerebellum	Left Hemisphere
		Flaccid	UUMN	Spastic	Hypokinetic	Hyperkinetic	Ataxic	Apraxia of Speech
afeafeafea								
Total Count X								
Total Count XX								
Total Count -								
Total Count of Yellow and Green Substracting Red		0	0	0	0	0	0	0